Doctor's Question Bank — Solved

نماذج أسئلة الدكتورة — بنك الأسئلة كاملاً مع الإجابات

Source: the doctor's question-bank PDF uploaded by the student (84 pages): True/False, Multiple choice and Short-answer questions per chapter, plus a compiled list and topic lists. The question wording is kept as in the file (including its typos).

- **509** unique questions in total after removing 91 exact duplicates (the same question appeared up to twice in the file; all duplicate answers agreed).
- **428** answers come from the doctor's file (red marks in the PDF, or the answer line printed before/after the question).
- **81** answers are **solved by Claude** because the file shows no red answer for them (compiled list Q54 and Q121–Q200). They are tagged and checked against Stallings' textbook wording; a few lower-confidence ones carry a ⚠ note.
- Chapter 2 Multiple-choice question #4 is empty in the source file, so it is omitted. Typos in short answers (e.g. "littels", "SPES", "PCL") were corrected to Little's, SPEC and PCI.
- **Original website:** I could not find a web page that matches these questions word for word. The chapter titles match William Stallings' *Computer Organization and Architecture* (10th ed.) and the format (T/F, MCQ, short answer per chapter) matches the publisher's instructor test bank that circulates on study sites, so the file is most likely a copy of that test bank (or a lecturer's adaptation). This is an inference, not a confirmed citation.

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Chapter 1: Basic Concepts and Computer Evolution

True / False — (15) صح / خطأ

1 **A computer is a complex system.**
Answer: **True**

2 **A computer organization does not need to be designed to implement a particular architectural specification.**
Answer: **False**

3 **Computer organization refers to attributes of a system visible to the programmer.**
Answer: **False**

4 **Changes in computer technology are finally slowing down.**
Answer: **False**

5 **Both the structure and functioning of a computer are, in essence, simple.**
Answer: **True**

6 **The number of bits used to represent various data types is an example of an architectural attribute.**
Answer: **True**

7 **Interfaces between the computer and peripherals is an example of an organizational attribute.**
Answer: **True**

8 **Historically the distinction between architecture and organization has not been an important one.**
Answer: **False**

9 **A particular architecture may span many years and encompass a number of different computer models, its organization changing with changing technology.**
Answer: **True**

10 **A microcomputer architecture and organization relationship is not very close.**
Answer: **False**

11 **Changes in technology not only influence organization but also result in the introduction of more powerful and more complex architectures.**
Answer: **True**

12 **The hierarchical nature of complex systems is essential to both their design and their description.**
Answer: **True**

13 **Both the structure and functioning of a computer are, in essence, simple.**
Answer: **True**

14 **A computer must be able to process, store, move, and control data.**
Answer: **True**

15 **When data are moved over longer distances, to or from a remote device, the process is known as data transport.**
Answer: **False**

Multiple Choice — (15) اختيار من متعدد

1 **Computer technology is changing at a _____ pace.**
A. slow
B. slow to medium
C. rapid ✓
D. non-existent
Answer: **C. rapid**

2 **Computer _____ refers to those attributes that have a direct impact on the logical execution of a program.**
A. organization
B. specifics
C. design
D. architecture ✓
Answer: **D. architecture**

3 **Architectural attributes include _____.**
A. I/O mechanisms ✓
B. control signals
C. interfaces
D. memory technology used
Answer: **A. I/O mechanisms**

4 **_____ attributes include hardware details transparent to the programmer.**
A. Interface
B. Organizational ✓
C. Memory
D. Architectural
Answer: **B. Organizational**

5 **It is a(n) _____ design issue whether a computer will have a multiply instruction.**
A. architectural ✓
B. memory
C. elementary
D. organizational
Answer: **A. architectural**

6 **It is a(n) _____ issue whether the multiply instruction will be implemented by a special multiply unit or by a mechanism that makes repeated use of the add unit of the system.**
A. architectural
B. memory
C. mechanical
D. organizational ✓
Answer: **D. organizational**

7 **A _____ system is a set of interrelated subsystems.**
A. secondary
B. hierarchical ✓
C. complex
D. functional
Answer: **B. hierarchical**

8 **An I/O device is referred to as a _____.**
A. CPU
B. control device
C. peripheral ✓
D. register
Answer: **C. peripheral**

9 **When data are moved over longer distances, to or from a remote device, the process is known as _____.**

A. data communications ✓
B. registering
C. structuring
D. data transport

Answer: **A. data communications**

10 **The _____ stores data.**

A. system bus
B. I/O
C. main memory ✓
D. control unit

Answer: **C. main memory**

11 **The _____ moves data between the computer and its external environment.**

A. data transport
B. I/O ✓
C. register
D. CPU interconnection

Answer: **B. I/O**

12 **A common example of system interconnection is by means of a _____.**

A. register
B. system bus ✓
C. data transport
D. control device

Answer: **B. system bus**

13 **A _____ is a mechanism that provides for communication among CPU, main memory, and I/O.**

A. system interconnection ✓
B. CPU interconnection
C. peripheral
D. processor

Answer: **A. system interconnection**

14 **_____ provide storage internal to the CPU.**

A. Control units
B. ALUs
C. Main memory
D. Registers ✓

Answer: **D. Registers**

15 **The _____ performs the computer's data processing functions.**

A. Register
B. CPU interconnection
C. ALU ✓
D. system bus

Answer: **C. ALU**

Fill in the blank — (15) أكمل الفراغ

1 **_____ refers to those attributes of a system visible to a programmer.**

Answer: **Computer architecture**

2 **_____ refers to the operational units and their interconnections that realize the architectural specifications.**

Answer: **Computer organization**

3 **Control signals, interfaces between the computer and peripherals, and the memory technology used are all examples of _____ attributes.**

Answer: **organizational**

4 **The instruction set, the number of bits used to represent various data types, I/O mechanisms and techniques for addressing memory are all examples of _____ attributes.**

Answer: **architectural**

5 **The _____ architecture is the architecture of IBM's mainframe product line.**

Answer: **System/370**

6 **_____ is the way in which the components are interrelated.**

Answer: **Structure**

7 **_____ is the operation of each individual component as part of the structure.**

Answer: **Function**

8 **The basic functions that a computer can perform are: data processing, data movement, control, and _____.**

Answer: **data storage**

9 **When data are received from or delivered to a device that is directly connected to the computer, the process is known as _____.**

Answer: **input/output (I/O)**

10 **The four main structural components of the computer are: main memory, I/O, system interconnection, and _____.**

Answer: **CPU**

11 **Often referred to as a processor, the _____ controls the operation of the computer and performs its data processing functions.**

Answer: **CPU**

12 **A common example of system interconnection is by means of a _____, consisting of a number of conducting wires to which all the other components attach.**

Answer: **system bus**

13 **The major structural components of the CPU are: control unit, register, CPU interconnection, and _____.**

Answer: **ALU**

14 **A _____ control unit operates by executing microinstructions that define the functionality of the control unit.**

Answer: **microprogrammed**

15 **The _____ controls the operation of the CPU and hence the computer.**

Answer: **control unit**

Chapter 2: Performance Issues

True / False — (15) صح / خطأ

1 **Year by year the cost of computer systems continues to rise.**
Answer: **False**

2 **Processors are so inexpensive that we now have microprocessors we throw away.**
Answer: **True**

3 **Workstation systems cannot support highly sophisticated engineering and scientific applications.**
Answer: **False**

4 **The IAS is the prototype of all subsequent general-purpose computers.**
Answer: **True**

5 **Cloud service providers use massive high-performance banks of for a servers to satisfy high-volume,high-transaction-rate applications broad spectrum of clients.**
Answer: **True**

6 **The raw speed of the microprocessor will not achieve its potential unless it is fed a constant stream of work to do in the form of computer instructions.**
Answer: **True**

7 **Superscalar execution is the same principle as seen in an assembly line.**
Answer: **False**

8 **Branch prediction potentially increases the amount of work available for the processor to execute.**
Answer: **True**

9 **Raw speed is far more important than how a processor performs when executing a given application.**
Answer: **False**

10 **The cache holds recently accessed data.**
Answer: **True**

11 **Operations performed by a processor, such as fetching an instruction, decoding the instruction, performing an arithmetic operation, and so on, are governed by a system clock.**
Answer: **True**

12 **A common measure of performance for a processor is the rate at which instructions are executed, expressed as millions of instructions per second (MIPS).**
Answer: **True**

13 **Designers wrestle with the challenge of balancing processor performance with that of main memory and other computer components.**
Answer: **True**

14 **A straight comparison of clock speeds on different processors tells the whole story about performance.**
Answer: **False**

15 **Measures such as MIPS and MFLOPS have proven adequate to evaluating the performanceof processors.**
Answer: **False**

Multiple Choice — (15) اختيار من متعدد

1 **Multiple parallel pipelines are used in _____.**
A. speculative execution
B. data flow analysis
C. superscalar execution ✓
D. branch prediction
Answer: **C. superscalar execution**

2 **The desktop application(s) that require the great power of today's microprocessor- based systems include_____.**
A. image processing
B. speech recognition
C. videoconferencing
D. all of the above ✓
Answer: **D. all of the above**

3 **_____ potentially increases the amount of work available for the processor to execute.**
A. Branch prediction ✓
B. Performance balance
C. Pipelining
D. BIPS
Answer: **A. Branch prediction**

4 **The interface between processor and _____ is the most crucial pathway in the entire computer because it is responsible for carrying a constant flow of program instructions and data between memory chips and the processor.**

A. main memory ✓
B. pipeline
C. clock speed
D. control unit

Answer: **A. main memory**

5 **The _____ is a relatively small fast memory interposed between a larger, slower memory and the logic that accesses the larger memory.**

A. peripheral
B. cache ✓
C. processor
D. arithmetic and logic unit

Answer: **B. cache**

6 **An increase in clock rate means that individual operations are executed ____.**

A. the same
B. slower
C. with very little change
D. more rapidly ✓

Answer: **D. more rapidly**

7 **A _____ is a core designed to perform parallel operations on graphics data.**

A. MIC
B. ALU
C. GPU ✓
D. PGD

Answer: **C. GPU**

8 **A(n) _____ Mean is a good candidate for comparing the execution time performance of several systems.**

A. Composite
B. Arithmetic ✓
C. Harmonic
D. Evaluation

Answer: **B. Arithmetic**

9 **_____ law deals with the potential speedup of a program using multiple processors compared to a single processor.**

A. Moore's
B. Amdahl's ✓
C. Little's
D. Murphy's

Answer: **B. Amdahl's**

10 **One increment, or pulse, of a clock is referred to as a _____.**

A. clock cycle ✓
B. clock rate
C. clock speed
D. cycle time

Answer: **A. clock cycle**

11 **The use of multiple processors on the same chip is referred to as _____ and provides the potential to increase performance without increasing the clock rate.**

A. multicore ✓
B. GPU
C. data channels
D. MPC

Answer: **A. multicore**

12 **With respect to changes in values, the _____ Mean gives equal weight to all of the values in the data set.**

A. Harmonic
B. Arithmetic
C. Composite
D. Geometric ✓

Answer: **D. Geometric**

13 **The _____ measures the ability of a computer to complete a single task.**

A. clock speed
B. speed metric ✓
C. execute cycle
D. cycle time

Answer: **B. speed metric**

14 **A measurement of how many tasks a computer can accomplish in a certain amount of time is called a(n) _____.**

A. real-time system
B. application analysis
C. cycle speed
D. throughput ✓

Answer: **D. throughput**

15 **The best known of the SPEC benchmark suites is _____.**

A. SPEC CPU2006 ✓
B. SPECjvm2008
C. SPECsfs2008
D. SPEC SC2013

Answer: **A. SPEC CPU2006**

Fill in the blank — (15) أكمل الفراغ

1 **_____ enables a processor to work simultaneously on multiple instructions by performing a different phase for each of the multiple instructions at the same time.**
Answer: **Pipelining**

2 **_____ is the ability to issue more than one instruction in every processor clock cycle.**
Answer: **Superscalar execution**

3 **_____: the processor looks ahead in the instruction code fetched from memory and predicts which branches, or groups of instructions, are likely to be processed next.**
Answer: **Branch prediction**

4 **_____ enables the processor to keep its execution engines as busy as possible by executing instructions that are likely to be needed.**
Answer: **Speculative execution**

5 **Traditionally found on a plug-in graphics card, a _____ is used to encode and render 2D and 3D graphics as well as process video.**
Answer: **GPU**

6 **_____ Law applies to a queuing system.**
Answer: **Little's**

7 **The three common formulas used for calculating a mean are arithmetic, harmonic, and _____.**
Answer: **geometric**

8 **The _____ Mean used for a time-based variable, such as program execution time, has the important property that it is directly proportional to the total time.**
Answer: **Arithmetic**

9 **The _____ Mean is preferred when calculating rates.**
Answer: **Harmonic**

10 **The _____ Mean gives consistent results regardless of which system is used as a reference.**
Answer: **geometric**

11 **_____ metrics are required for all reported results and have strict guidelines for compilation.**
Answer: **Base**

12 **A _____ suite is a collection of programs, defined in a high-level language, that together attempt to provide a representative test of a computer in a particular application or system programming area.**
Answer: **benchmark**

13 **At the most fundamental level, the speed of a processor is dictated by the pulse frequency produced by the clock, measured in cycles per second, or _____.**
Answer: **Hertz**

14 **The best-known collection of benchmark suites is defined and maintained by an industry consortium known as _____.**
Answer: **SPEC**

15 **_____ law deals with the potential speedup of a program using multiple processors compared to a single processor.**
Answer: **Amdahl's**

Chapter 3: A Top-Level View of Computer Function and Interconnection

True / False — (15) صح / خطأ

1 **At a top level, a computer consists of CPU, memory, and I/O components.**
Answer: **True**

2 **The basic function of a computer is to execute programs.**
Answer: **True**

3 **Program execution consists of repeating the process of instruction fetch and instruction execution.**
Answer: **True**

4 **Interrupts do not improve processing efficiency.**
Answer: **False**

5 **An I/O module cannot exchange data directly with the processor.**
Answer: **False**

6 **A key characteristic of a bus is that it is not a shared transmission medium.**
Answer: **False**

7 **Computer systems contain a number of different buses that provide pathways between components at various levels of the computer system hierarchy.**
Answer: **True**

8 **In general, the more devices attached to the bus, the greater the bus length and hence the greater the propagation delay.**
Answer: **True**

9 **It is not possible to connect I/O controllers directly onto the system bus.**
Answer: **False**

10 **The method of using the same lines for multiple purposes is known as time multiplexing.**
Answer: **True**

11 **Timing refers to the way in which events are coordinated on the bus.**
Answer: **True**

12 **With asynchronous timing the occurrence of events on the bus is determined by a clock.**
Answer: **False**

13 **Because all devices on a synchronous bus are tied to a fixed clock rate, the system cannot take advantage of advances in device performance.**
Answer: **True**

14 **The unit of transfer at the link layer is a phit and the unit transfer at the physical layer is a flit.**
Answer: **False**

15 **A key requirement for PCIe is high capacity to support the needs ofhigher data rate I/O devices such as Gigabit Ethernet.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **Virtually all contemporary computer designs are based on concepts developed by _____ at the Institute for Advanced Studies, Princeton.**
A. John Maulchy
B. John von Neumann ✓
C. Herman Hollerith
D. John Eckert
Answer: **B. John von Neumann**

2 **The von Neumann architecture is based on which concept?**
A. data and instructions are stored in a single read-write memory
B. the contents of this memory are addressable by location
C. execution occurs in a sequential fashion
D. all of the above ✓
Answer: **D. all of the above**

3 **A sequence of codes or instructions is called _____.**
A. software ✓
B. memory
C. an interconnect
D. a register
Answer: **A. software**

4 **The processing required for a single instruction is called a(n) _____ cycle.**
A. execute
B. fetch
C. instruction ✓
D. packet
Answer: **C. instruction**

5 **A(n) _____ is generated by a failure such as power failure or memory parity error.**

A. I/O interrupt
B. hardware failure interrupt ✓
C. timer interrupt
D. program interrupt

Answer: **B. hardware failure interrupt**

6 **A(n) _____ is generated by some condition that occurs as a result of an instruction execution.**

A. timer interrupt
B. I/O interrupt
C. program interrupt ✓
D. hardware failure interrupt

Answer: **C. program interrupt**

7 **The interconnection structure must support which transfer?**

A. memory to processor
B. processor to memory
C. I/O to or from memory
D. all of the above ✓

Answer: **D. all of the above**

8 **A bus that connects major computer components (processor, memory, I/O) is called a _____.**

A. system bus ✓
B. address bus
C. data bus
D. control bus

Answer: **A. system bus**

9 **The _____ are used to designate the source or destination of the data on the data bus.**

A. system lines
B. data lines
C. control lines
D. address lines ✓

Answer: **D. address lines**

10 **The data lines provide a path for moving data among system modules and are collectively called the _____.**

A. control bus
B. address bus
C. data bus ✓
D. system bus

Answer: **C. data bus**

11 **A _____ is the high-level set of rules for exchanging packets of data between devices.**

A. bus
B. protocol ✓
C. packet
D. QPI

Answer: **B. protocol**

12 **Each data path consists of a pair of wires (referred to as a _____) that transmits data one bit at a time.**

A. lane ✓
B. path
C. line
D. bus

Answer: **A. lane**

13 **The _____ receives read and write requests from the software above the TL and creates request packets for transmission to a destination via the link layer.**

A. transaction layer ✓
B. root layer
C. configuration layer
D. transport layer

Answer: **A. transaction layer**

14 **The TL supports which of the following address spaces?**

A. memory
B. I/O
C. message
D. all of the above ✓

Answer: **D. all of the above**

15 **The QPI _____ layer is used to determine the course that a packet will traverse across the available system interconnects.**

A. link
B. protocol
C. routing ✓
D. physical

Answer: **C. routing**

Fill in the blank — (15) أكمل الفراغ

1 **A ____ register specifies the address in memory for the next read or write.**

Answer: **memory address (MAR)**

2 **A ____ register contains the data to be written into memory or receives the data read from memory.**

Answer: **memory buffer (MBR)**

3 **The most common classes of interrupts are: program, timer, I/O and _____.**
Answer: **hardware failure**

4 **A(n) _____ interrupt is generated by a timer within the processor and allows the operating system to perform certain functions on a regular basis.**
Answer: **timer**

5 **A(n) _____ interrupt is generated by an I/O controller to signal normal completion of an operation, request service from the processor, or to signal a variety of error conditions.**
Answer: **I/O**

6 **A _____ interrupt simply means that the processor can and will ignore that interrupt request signal.**
Answer: **disabled**

7 **The collection of paths connecting the various modules is called the _____ structure.**
Answer: **interconnection**

8 **A _____ is a communication pathway connecting two or more devices.**
Answer: **bus**

9 **The _____ lines are used to control the access to and the use of the data and address lines.**
Answer: **control**

10 **There are three important groups of DLLPs used in managing a link: flow control packets, _____, and TLP ACK and NAK packets.**
Answer: **power management packets**

11 **The purpose of the PCIe _____ layer is to ensure reliable delivery of packets across the PCIe link.**
Answer: **data link**

12 **With _____ transmission, signals are transmitted as a current that travels down one conductor and returns on the other.**
Answer: **balanced**

13 **The QPI link layer performs two key functions: flow control and _____ control.**
Answer: **error**

14 **The _____ is a popular high-bandwidth, processor-independent bus that can function as a mezzanine or peripheral bus.**
Answer: **PCI**

15 **The _____ function is needed to ensure that a sending QPI entity does not overwhelm a receiving QPI entity by sending data faster than the receiver can process the data and clear buffers for more incoming data.**
Answer: **flow control**

Chapter 4: Cache Memory & Memory Characteristics

True / False — (15) صح / خطأ

1 **No single technology is optimal in satisfying the memory requirements for a computer system.**
Answer: **True**

2 **A typical computer system is equipped with a hierarchy of memory subsystems, some internal to the system and some external.**
Answer: **True**

3 **External memory is often equated with main memory.**
Answer: **False**

4 **The processor requires its own local memory.**
Answer: **True**

5 **Cache is not a form of internal memory.**
Answer: **False**

6 **The unit of transfer must equal a word or an addressable unit.**
Answer: **False**

7 **Both sequential access and direct access involve a shared read- write mechanism.**
Answer: **True**

8 **In a volatile memory, information decays naturally or is lost when electrical power is switched off.**
Answer: **True**

9 **To achieve greatest performance the memory must be able to keep up with the processor.**
Answer: **True**

10 **Secondary memory is used to store program and data files and is usually visible to the programmer only in terms of individual bytes or words.**
Answer: **False**

11 **The L1 cache is slower than the L3 cache.**
Answer: **False**

12 **With write back updates are made only in the cache.**
Answer: **True**

13 **It has become possible to have a cache on the same chip as the processor.**
Answer: **True**

14 **All of the Pentium processors include two on-chip L1 caches, one for data and one for instructions.**
Answer: **True**

15 **Cache design for HPC is the same as that for other hardware platforms and applications.**
Answer: **False**

Multiple Choice — (15) اختيار من متعدد

1 **_____ refers to whether memory is internal or external to the computer.**
A. Location ✓
B. Access
C. Hierarchy
D. Tag
Answer: **A. Location**

2 **Internal memory capacity is typically expressed in terms of _____.**
A. hertz
B. nanos
C. bytes ✓
D. LOR
Answer: **C. bytes**

3 **For internal memory, the _____ is equal to the number of electrical lines into and out of the memory module.**
A. access time
B. unit of transfer ✓
C. capacity
D. memory ratio
Answer: **B. unit of transfer**

4 **“Memory is organized into records and access must be made in a specific linear sequence” is a description of _____.**
A. sequential access ✓
B. direct access
C. random access
D. associative access
Answer: **A. sequential access**

5 **individual blocks or records have a unique address based on physical location with _____.**

A. associative
B. physical access access
C. direct access ✓
D. sequential access

Answer: **C. direct access**

6 **For random-access memory, _____ is the time from the instant that an address is presented to the memory to the instant that data have been stored or made available for use.**

A. memory cycle time
B. direct access
C. transfer rate
D. access time ✓

Answer: **D. access time**

7 **The _____ consists of the access time plus any additional time required before a second access can commence.**

A. latency
B. memory cycle time ✓
C. direct access
D. transfer rate

Answer: **B. memory cycle time**

8 **A portion of main memory used as a buffer to hold data temporarily that is to be read out to disk is referred to as a _____.**

A. disk cache ✓
B. latency
C. virtual address
D. miss

Answer: **A. disk cache**

9 **A line includes a _____ that identifies which particular block is currently being stored.**

A. cache
B. hit
C. tag ✓
D. locality

Answer: **C. tag**

10 **_____ is the simplest mapping technique and maps each block of main memory into only one possible cache line.**

A. Direct mapping ✓
B. Associative mapping
C. Set associative mapping
D. None of the above

Answer: **A. Direct mapping**

11 **When using the _____ technique all write operations made to main memory are made to the cache as well.**

A. write back
B. LRU
C. write through ✓
D. unified cache

Answer: **C. write through**

12 **The key advantage of the _____ design is that it eliminates contention for the cache between the instruction fetch/decode unit and the execution unit.**

A. logical cache
B. split cache ✓
C. unified cache
D. physical cache

Answer: **B. split cache**

13 **The Pentium 4 _____ component executes micro-operations, fetching the required data from the L1 data cache and temporarily storing results in registers.**

A. fetch/decode unit
B. out-of-order execution logic
C. execution unit ✓
D. memory subsystem

Answer: **C. execution unit**

14 **In reference to access time to a two-level memory, a _____ occurs if an accessed word is not found in the faster memory.**

A. miss ✓
B. hit
C. line
D. tag

Answer: **A. miss**

15 **A logical cache stores data using _____.**

A. physical addresses
B. virtual addresses ✓
C. random addresses
D. none of the above

Answer: **B. virtual addresses**

Fill in the blank — (15) أكمل الفراغ

1 **_____ memory consists of peripheral storage devices, such as disk and tape.**

Answer: **External**

2 **One byte equals _____ bits.**

Answer: **8**

3 **From a user's point of view the two most important characteristics of memory are capacity and _____.**

Answer: **performance**

4 **The three performance parameters for memory are: access time, transfer rate, and _____.**

Answer: **memory cycle time**

5 **_____ is a random access type of memory that enables one to make a comparison of desired bit locations within a word for a specified match, and to do this for all words simultaneously, thus retrieving a word based on a portion of its contents rather than its address.**

Answer: **Associative memory**

6 **The _____ rate is the rate at which data can be transferred into or out of a memory unit.**

Answer: **transfer**

7 **The most commonly used physical types of memory are: semiconductor memory, _____ memory (used for disk and tape), and optical and magneto-optical.**

Answer: **magnetic surface**

8 **The three key characteristics of memory are capacity, access time, and _____.**

Answer: **cost**

9 **External, nonvolatile memory is referred to as _____ or auxiliary memory.**

Answer: **secondary**

10 **The cache consists of blocks called _____.**

Answer: **lines**

11 **_____ computing deals with super computers and their software.**

Answer: **High-performance**

12 **The Pentium 4 processor core consists of four major components: fetch/decode unit, out-of-order execution logic, memory subsystem, and _____.**

Answer: **execution units**

13 **The _____ units execute micro-operations, fetching the required data from the L1 data cache and temporarily storing results in registers.**

Answer: **execution**

14 **_____ memory is a facility that allows programs to address memory from a logical point of view, without regard to the amount of main memory physically available.**

Answer: **Virtual**

15 **For set-associative mapping the cache control logic interprets a memory address as three fields: Set, Word, and _____.**

Answer: **Tag**

Additional questions from the compiled list (Chapters 1-4 topics)

True / False — (16) صح / خطأ

1 **There is a tremendous variety of products, from single-chip microcomputers costing a few dollars to supercomputers costing tens of millions of dollars that can rightly claim the name "computer".**
Answer: **True**

2 **The variety of computer products is exhibited only in cost.**
Answer: **True**

3 **The textbook for this course is about the structure and function of computers.**
Answer: **True**

4 **The world's first general-purpose electronic digital computer was designed and constructed at The Ohio State University.**
Answer: **True**

5 **John Mauchly and John Eckert designed the ENIAC.**
Answer: **True**

6 **The major drawback of the EDVAC was that it had to be programmed manually by setting switches and plugging and unplugging cables.**
Answer: **False**

7 **The IAS operates by repetitively performing an instruction cycle.**
Answer: **True**

8 **Backward compatible means that the programs written for the older machines can be executed on the new machine.**
Answer: **True**

9 **A vacuum tube is a solid-state device made from silicon.**
Answer: **False**

10 **Computers are classified into generations based on the fundamental hardware technology employed.**
Answer: **True**

11 **System software was introduced in the third generation of computers.**
Answer: **False**

12 **A wafer is made of silicon and is broken up into chips which consists of many gates and/or memory cells plus a number of input and output attachment points.**
Answer: **True**

13 **IBM's System/360 was the industry's first planned family of computers.**
Answer: **True**

14 **Intel's 4004 was the first chip to contain all of the components of a CPU on a single chip.**
Answer: **True**

15 **The Intel x86 evolved from RISC design principles and is used in embedded systems.**
Answer: **False**

16 **A common measure of performance for a processor is the rate at which instructions are executed, expressed as billions of instructions per seconds (BIPS).**
Answer: **False**

Multiple Choice — (13) اختيار من متعدد

1 **The _____ was the world's first general-purpose electronic digital computer.**
A. UNIVAC
B. MARK IV
C. ENIAC ✓
D. Hollerith's Counting Machine
Answer: **C. ENIAC**

2 **The Electronic Numerical Integrator and Computer project was a response to U.S. needs during _____.**
A. the Civil War
B. the French-American War
C. World War I
D. World War II ✓
Answer: **D. World War II**

3 **The ENIAC used _____.**
A. vacuum tubes ✓
B. integrated circuits
C. IAS
Answer: **A. vacuum tubes**

4 **The ENIAC is an example of a _____ generation computer.**
A. first ✓
B. second
C. third
D. fourth
Answer: **A. first**

5 **The _____ interprets the instructions in memory and causes them to be executed.**

A. main memory
B. control unit ✓
C. I/O
D. arithmetic and logic unit

Answer: **B. control unit**

6 **The memory of the IAS consists of 1000 storage locations called _____.**

A. opcodes
B. wafers
C. VLSIs
D. words ✓

Answer: **D. words**

7 **The _____ contains the 8-bit opcode instruction being executed.**

A. memory buffer register
B. instruction buffer register
C. instruction register ✓
D. memory address register

Answer: **C. instruction register**

8 **During the _____ the opcode of the next instruction is loaded into the IR and the address portion is loaded into the MAR.**

A. executes cycle
B. fetch cycle ✓
C. instruction cycle
D. clock cycle

Answer: **B. fetch cycle**

9 **Second generation computers used _____.**

A. integrated circuits
B. Transistors ✓
C. vacuum tubes
D. large-scale integration

Answer: **B. Transistors**

answer solved by Claude — not marked in the source file

10 **The _____ defines the third generation of computers.**

A. integrated circuit ✓
B. vacuum tube
C. transistor
D. VLSI

Answer: **A. integrated circuit**

11 **With the _____, Intel introduced the use of superscalar techniques that allow multiple instructions to execute in parallel.**

A. Core
B. 8080
C. 80486
D. Pentium ✓

Answer: **D. Pentium**

12 **ARM processors are designed to meet the needs of _____.**

A. embedded real-time systems
B. application platforms
C. secure applications
D. all of the above ✓

Answer: **D. all of the above**

13 **One increment, or pulse, of the system clock is referred to as a _____.**

A. clock tick ✓
B. cycle time
C. clock rate
D. cycle speed

Answer: **A. clock tick**

Internal Memory (semiconductor memory)

True / False — (15) صح / خطأ

1 **The basic element of a semiconductor memory is the memory cell.**
Answer: **True**
answer solved by Claude — not marked in the source file

2 **A characteristic of ROM is that it is volatile.**
Answer: **False**
answer solved by Claude — not marked in the source file

3 **RAM must be provided with a constant power supply.**
Answer: **True**
answer solved by Claude — not marked in the source file

4 **The two traditional forms of RAM used in computers are DRAM and SRAM.**
Answer: **True**
answer solved by Claude — not marked in the source file

5 **A static RAM will hold its data as long as power is supplied to it.**
Answer: **True**
answer solved by Claude — not marked in the source file

6 **Nonvolatile means that power must be continuously supplied to the memory to preserve the bit values.**
Answer: **False**
answer solved by Claude — not marked in the source file

7 **The advantage of RAM is that the data or program is permanently in main memory and need never be loaded from a secondary storage device.**
Answer: **False**
answer solved by Claude — not marked in the source file

8 **Semiconductor memory comes in packaged chips.**
Answer: **True**
answer solved by Claude — not marked in the source file

9 **All DRAMs require a refresh operation.**
Answer: **True**
answer solved by Claude — not marked in the source file

10 **A number of chips can be grouped together to form a memory bank.**
Answer: **True**
answer solved by Claude — not marked in the source file

11 **An error-correcting code enhances the reliability of the memory at the cost of added complexity.**
Answer: **True**
answer solved by Claude — not marked in the source file

12 **DRAM is much costlier than SRAM.**
Answer: **False**
answer solved by Claude — not marked in the source file

13 **RDRAM is limited by the fact that it can only send data to the processor once per bus clock cycle.**
Answer: **False**
answer solved by Claude — not marked in the source file
⚠ الجملة تنسب القيد إلى RDRAM بينما المصدر المعروف (Stallings) ينسبه إلى SDRAM؛ اخترنا FALSE.

14 **The prefetch buffer is a memory cache located on the RAM chip.**
Answer: **True**
answer solved by Claude — not marked in the source file

15 **The SRAM on the CDRAM cannot be used as a buffer to support the serial access of a block of data.**
Answer: **False**
answer solved by Claude — not marked in the source file

Multiple Choice — (15) اختيار من متعدد

1 **Which properties do all semiconductor memory cells share?**
A. they exhibit two stable states which can be used to represent binary 1 and 0
B. they are capable of being written into to set the state
C. they are capable of being read to sense the state
D. all of the above ✓
Answer: **D. all of the above**
answer solved by Claude — not marked in the source file

2 **One distinguishing characteristic of memory that is designated as _____ is that it is possible to both to read data from the memory and to write new data into the memory easily and rapidly.**
A. RAM ✓
B. ROM
C. EPROM
D. EEPROM
Answer: **A. RAM**
answer solved by Claude — not marked in the source file

3 **Which of the following memory types are nonvolatile?**

A. erasable PROM
B. programmable ROM
C. flash memory
D. all of the above ✓

Answer: **D. all of the above**

answer solved by Claude — not marked in the source file

4 **In a _____, binary values are stored using traditional flip-flop logic-gate configurations.**

A. ROM
B. SRAM ✓
C. DRAM
D. RAM

Answer: **B. SRAM**

answer solved by Claude — not marked in the source file

5 **A _____ contains a permanent pattern of data that cannot be changed, is nonvolatile, and cannot have new data written into it.**

A. RAM
B. SRAM
C. ROM ✓
D. flash memory

Answer: **C. ROM**

answer solved by Claude — not marked in the source file

6 **With _____ the microchip is organized so that a section of memory cells are erased in a single action.**

A. flash memory ✓
B. SDRAM
C. DRAM
D. EEPROM

Answer: **A. flash memory**

answer solved by Claude — not marked in the source file

7 **_____ can be caused by harsh environmental abuse, manufacturing defects, and wear.**

A. SEC errors
B. Hard errors ✓
C. Syndrome errors
D. Soft errors

Answer: **B. Hard errors**

answer solved by Claude — not marked in the source file

8 **_____ can be caused by power supply problems or alpha particles.**

A. Soft errors ✓
B. AGT errors
C. Hard errors
D. SEC errors

Answer: **A. Soft errors**

answer solved by Claude — not marked in the source file

9 **The _____ exchanges data with the processor synchronized to an external clock signal and running at the full speed of the processor/memory bus without imposing wait states.**

A. DDR-DRAM
B. SDRAM ✓
C. CDRAM
D. none of the above

Answer: **B. SDRAM**

answer solved by Claude — not marked in the source file

10 **_____ can send data to the processor twice per clock cycle.**

A. CDRAM
B. SDRAM
C. DDR-DRAM ✓
D. RDRAM

Answer: **C. DDR-DRAM**

answer solved by Claude — not marked in the source file

11 **_____ increases the data transfer rate by increasing the operational frequency of the RAM chip and by increasing the prefetch buffer from 2 bits to 4 bits per chip.**

A. DDR2 ✓
B. RDRAM
C. CDRAM
D. DDR3

Answer: **A. DDR2**

answer solved by Claude — not marked in the source file

12 **_____ increases the prefetch buffer size to 8 bits.**

A. CDRAM
B. RDRAM
C. DDR3 ✓
D. all of the above

Answer: **C. DDR3**

answer solved by Claude — not marked in the source file

13 **Theoretically, a DDR module can transfer data at a clock rate in the range of _____ MHz.**

A. 200 to 600 ✓
B. 400 to 1066
C. 600 to 1400
D. 800 to 1600

Answer: **A. 200 to 600**

answer solved by Claude — not marked in the source file

> ⚠ حسب Stallings: DDR من 200 إلى 600 MHz (DDR2: 400–800، DDR3: 800–1600).

14 **A DDR3 module transfers data at a clock rate of _____ MHz.**

A. 600 to 1200
B. 800 to 1600 ✓
C. 1000 to 2000
D. 1500 to 3000

Answer: **B. 800 to 1600**

answer solved by Claude — not marked in the source file

15 **The _____ enables the RAM chip to preposition bits to be placed on the data bus as rapidly as possible.**

A. flash memory
B. Hamming code
C. RamBus
D. buffer ✓

Answer: **D. buffer**

answer solved by Claude — not marked in the source file

External Memory (disks, RAID, SSD, optical)

True / False — (15) صح / خطأ

1 **Magnetic disks are the foundation of external memory on virtually all computer systems.**
Answer: **True**
answer solved by Claude — not marked in the source file

2 **During a read or write operation, the head rotates while the platter beneath it stays stationary.**
Answer: **False**
answer solved by Claude — not marked in the source file

3 **The width of a track is double that of the head.**
Answer: **False**
answer solved by Claude — not marked in the source file

4 **There are typically hundreds of sectors per track and they may be either fixed or variable lengths.**
Answer: **True**
answer solved by Claude — not marked in the source file

5 **A bit near the center of a rotating disk travels past a fixed point slower than a bit on the outside.**
Answer: **True**
answer solved by Claude — not marked in the source file

6 **The disadvantage of using CAV is that individual blocks of data can only be directly addressed by track and sector.**
Answer: **False**
answer solved by Claude — not marked in the source file

7 **A removable disk can be removed and replaced with another disk.**
Answer: **True**
answer solved by Claude — not marked in the source file

8 **The head must generate or sense an electromagnetic field of sufficient magnitude to write and read properly.**
Answer: **True**
answer solved by Claude — not marked in the source file

9 **The transfer time to or from the disk does not depend on the rotation speed of the disk.**
Answer: **False**
answer solved by Claude — not marked in the source file

10 **RAID is a set of physical disk drives viewed by the operating system as a single logical drive.**
Answer: **True**
answer solved by Claude — not marked in the source file

11 **RAID level 0 is not a true member of the RAID family because it does not include redundancy to improve performance.**
Answer: **True**
answer solved by Claude — not marked in the source file

> ⚠ الصياغة «redundancy to improve performance» كما في كتاب Stallings؛ اخترنا TRUE.

12 **Because data are striped in very small strips, RAID 3 cannot achieve very high data transfer rates.**
Answer: **False**
answer solved by Claude — not marked in the source file

13 **The SSDs now on the market use a type of semiconductor memory referred to as flash memory.**
Answer: **True**
answer solved by Claude — not marked in the source file

14 **SSD performance has a tendency to speed up as the device is used.**
Answer: **False**
answer solved by Claude — not marked in the source file

15 **Flash memory becomes unusable after a certain number of writes.**
Answer: **True**
answer solved by Claude — not marked in the source file

Multiple Choice — (15) اختيار من متعدد

1 **Greater ability to withstand shock and damage, improvement in the uniformity of the magnet film surface to increase disk reliability, and a significant reduction in overall surface defects to help reduce read-write errors, are all benefits of _____.**

A. magnetic read and write mechanisms
B. platters
C. the glass substrate ✓
D. a solid state drive

Answer: **C. the glass substrate**

answer solved by Claude — not marked in the source file

2 **Adjacent tracks are separated by _____.**

A. sectors
B. gaps ✓
C. pits
D. heads

Answer: **B. gaps**

answer solved by Claude — not marked in the source file

3 **Data are transferred to and from the disk in _____.**

A. tracks
B. gaps
C. sectors ✓
D. pits

Answer: **C. sectors**

answer solved by Claude — not marked in the source file

4 **In most contemporary systems fixed-length sectors are used, with _____ bytes being the nearly universal sector size.**

A. 64
B. 128
C. 256
D. 512 ✓

Answer: **D. 512**

answer solved by Claude — not marked in the source file

5 **Scanning information at the same rate by rotating the disk at a fixed speed is known as the _____.**

A. constant angular velocity ✓
B. magnetoresistive
C. rotational delay
D. constant linear velocity

Answer: **A. constant angular velocity**

answer solved by Claude — not marked in the source file

6 **The disadvantage of _____ is that the amount of data that can be stored on the long outer tracks is only the same as what can be stored on the short inner tracks.**

A. SSD
B. CAV ✓
C. ROM
D. CLV

Answer: **B. CAV**

answer solved by Claude — not marked in the source file

7 **A _____ disk is permanently mounted in the disk drive, such as the hard disk in a personal computer.**

A. nonremovable ✓
B. movable-head
C. double sided
D. removable

Answer: **A. nonremovable**

answer solved by Claude — not marked in the source file

8 **When the magnetizable coating is applied to both sides of the platter the disk is then referred to as _____.**

A. multiple sided
B. substrate
C. double sided ✓
D. all of the above

Answer: **C. double sided**

answer solved by Claude — not marked in the source file

9 **The set of all the tracks in the same relative position on the platter is referred to as a _____.**

A. floppy disk
B. single-sided disk
C. sector
D. cylinder ✓

Answer: **D. cylinder**

answer solved by Claude — not marked in the source file

10 **The sum of the seek time and the rotational delay equals the _____, which is the time it takes to get into position to read or write.**

A. access time ✓
B. gap time
C. transfer time
D. constant angular velocity

Answer: **A. access time**

answer solved by Claude — not marked in the source file

11 **_____ is the standardized scheme for multiple-disk database design.**

A. RAID ✓
B. CAV
C. CLV
D. SSD

Answer: **A. RAID**

answer solved by Claude — not marked in the source file

12 **RAID level _____ has the highest disk overhead of all RAID types.**

A. 0
B. 1 ✓
C. 3
D. 5

Answer: **B. 1**

answer solved by Claude — not marked in the source file

13 **A _____ is a high-definition video disk that can store 25 Gbytes on a single layer on a single side.**

A. DVD
B. DVD-R
C. DVD-RW
D. Blu-ray DVD ✓

Answer: **D. Blu-ray DVD**

answer solved by Claude — not marked in the source file

14 **_____ is when the disk rotates more slowly for accesses near the outer edge than for those near the center.**

A. Constant angular velocity (CAV)
B. Magnetoresistive
C. Constant linear velocity (CLV) ✓
D. Seek time

Answer: **C. Constant linear velocity (CLV)**

answer solved by Claude — not marked in the source file

15 **The areas between pits are called _____.**

A. lands ✓
B. sectors
C. cylinders
D. strips

Answer: **A. lands**

answer solved by Claude — not marked in the source file

Input/Output (I/O)

True / False — (15) صح / خطأ

1 **A set of I/O modules is a key element of a computer system.**
Answer: **True**
answer solved by Claude — not marked in the source file

2 **An I/O module must recognize one unique address for each peripheral it controls.**
Answer: **True**
answer solved by Claude — not marked in the source file

3 **I/O channels are commonly seen on microcomputers, whereas I/O controllers are used on mainframes.**
Answer: **False**
answer solved by Claude — not marked in the source file

4 **It is the responsibility of the processor to periodically check the status of the I/O module until it finds that the operation is complete.**
Answer: **True**
answer solved by Claude — not marked in the source file

5 **With isolated I/O there is a single address space for memory locations and I/O devices.**
Answer: **False**
answer solved by Claude — not marked in the source file

6 **A disadvantage of memory-mapped I/O is that valuable memory address space is used up.**
Answer: **True**
answer solved by Claude — not marked in the source file

7 **The disadvantage of the software poll is that it is time consuming.**
Answer: **True**
answer solved by Claude — not marked in the source file

8 **With a daisy chain the processor just picks the interrupt line with the highest priority.**
Answer: **False**
answer solved by Claude — not marked in the source file

9 **Bus arbitration makes use of vectored interrupts.**
Answer: **True**
answer solved by Claude — not marked in the source file

> ⚠ حسب Stallings: bus arbitration شكل من أشكال vectored interrupt.

10 **The rotating interrupt mode allows the processor to inhibit interrupts from certain devices.**
Answer: **False**
answer solved by Claude — not marked in the source file

11 **Because the 82C55A is programmable via the control register, it can be used to control a variety of simple peripheral devices.**
Answer: **True**
answer solved by Claude — not marked in the source file

12 **When large volumes of data are to be moved, a more efficient technique is direct memory access (DMA).**
Answer: **True**
answer solved by Claude — not marked in the source file

13 **An I/O channel has the ability to execute I/O instructions, which gives it complete control over I/O operations.**
Answer: **True**
answer solved by Claude — not marked in the source file

14 **A multipoint external interface provides a dedicated line between the I/O module and the external device.**
Answer: **False**
answer solved by Claude — not marked in the source file

15 **A Thunderbolt compatible peripheral interface is no more complex than that of a simple USB device.**
Answer: **False**
answer solved by Claude — not marked in the source file

Multiple Choice — (15) اختيار من متعدد

1 **The _____ contains logic for performing a communication function between the peripheral and the bus.**
A. I/O channel
B. I/O module ✓
C. I/O processor
D. I/O command
Answer: **B. I/O module**
answer solved by Claude — not marked in the source file

2 **The most common means of computer/user interaction is a _____.**
A. keyboard/monitor ✓
B. mouse/printer
C. modem/printer
D. monitor/printer
Answer: **A. keyboard/monitor**
answer solved by Claude — not marked in the source file

3 **The I/O function includes a _____ requirement to coordinate the flow of traffic between internal resources and external devices.**

A. cycle
B. status reporting
C. control and timing ✓
D. data

Answer: **C. control and timing**

answer solved by Claude — not marked in the source file

4 **An I/O module that takes on most of the detailed processing burden, presenting a high-level interface to the processor, is usually referred to as an _____.**

A. I/O channel ✓
B. I/O command
C. I/O controller
D. device controller

Answer: **A. I/O channel**

answer solved by Claude — not marked in the source file

> ⚠ حسب Stallings: «I/O channel» أو «I/O processor»؛ الخيار المتاح هو I/O channel.

5 **An I/O module that is quite primitive and requires detailed control is usually referred to as an _____.**

A. I/O command
B. I/O controller ✓
C. I/O channel
D. I/O processor

Answer: **B. I/O controller**

answer solved by Claude — not marked in the source file

6 **The _____ command causes the I/O module to take an item of data from the data bus and subsequently transmit that data item to the peripheral.**

A. control
B. test
C. read
D. write ✓

Answer: **D. write**

7 **The _____ command is used to activate a peripheral and tell it what to do.**

A. control ✓
B. test
C. read
D. write

Answer: **A. control**

8 **_____ is when the DMA module must force the processor to suspend operation temporarily.**

A. Interrupt
B. Thunderbolt
C. Cycle stealing ✓
D. Lock down

Answer: **C. Cycle stealing**

9 **The 8237 DMA is known as a _____ DMA controller.**

A. command
B. cycle stealing
C. interrupt
D. fly-by ✓

Answer: **D. fly-by**

10 **_____ is a digital display interface standard now widely adopted for computer monitors, laptop displays, and other graphics and video interfaces.**

A. DisplayPort ✓
B. PCI Express
C. Thunderbolt
D. InfiniBand

Answer: **A. DisplayPort**

11 **The _____ layer is the key to the operation of Thunderbolt and what makes it attractive as a high-speed peripheral I/O technology.**

A. cable
B. application
C. common transport ✓
D. physical

Answer: **C. common transport**

12 **The Thunderbolt protocol _____ layer is responsible for link maintenance including hot-plug detection and data encoding to provide highly efficient data transfer.**

A. cable
B. application
C. common transport
D. physical ✓

Answer: **D. physical**

13 **The _____ contains I/O protocols that are mapped on to the transport layer.**

A. cable
B. application ✓
C. common transport
D. physical

Answer: **B. application**

14 **A _____ is used to connect storage systems, routers, and other peripheral devices to an InfiniBand switch.**

A. target channel adapter ✓
B. InfiniBand switch
C. host channel adapter
D. subnet

Answer: **A. target channel adapter**

15 **A _____ connects InfiniBand subnets, or connects an InfiniBand switch to a network such as a local area network, wide area network, or storage area network.**

A. memory controller
B. TCA
C. HCA
D. router ✓

Answer: **D. router**

Operating Systems

True / False — (15) صح / خطأ

1 **Scheduling and memory management are the two OS functions that are most relevant to the study of computer organization and architecture.**
Answer: **True**

2 **The end user is concerned mainly with the computer's architecture.**
Answer: **False**

3 **The most important system program is the OS.**
Answer: **True**

4 **The ABI is the boundary between hardware and software.**
Answer: **False**

5 **The OS must determine how much processor time is to be devoted to the execution of a particular user program.**
Answer: **True**

6 **With a batch operating system the user does not have direct access to the processor.**
Answer: **True**

7 **Privileged instructions are certain instructions that are designated special and can be executed only by the monitor.**
Answer: **True**

8 **Uniprogramming is the central theme of modern operating systems.**
Answer: **False**

9 **Both batch multiprogramming and time sharing use multiprogramming.**
Answer: **True**

10 **An interrupt is a hardware-generated signal to the processor.**
Answer: **True**

11 **Swapping is an I/O operation.**
Answer: **True**

12 **With demand paging it is necessary to load an entire process into main memory.**
Answer: **False**

13 **The Pentium II includes hardware for both segmentation and paging.**
Answer: **True**

14 **ARM provides a versatile virtual memory system architecture that can be tailored to the needs of the embedded system designer.**
Answer: **True**

15 **Managers are users of domains that must observe the access permissions of the individual sections and/or pages that make up that domain.**
Answer: **False**

Multiple Choice — (15) اختيار من متعدد

1 **The _____ is a program that controls the execution of application programs and acts as an interface between applications and the computer hardware.**
A. job control language
B. operating system ✓
C. batch system
D. nucleus
Answer: **B. operating system**

2 **Facilities and services provided by the OS that assist the programmer in creating programs are in the form of _____ programs that are not actually part of the OS but are accessible through the OS.**
A. utility ✓
B. multitasking
C. JCL
D. logical address
Answer: **A. utility**

3 **The _____ defines the repertoire of machine language instructions that a computer can follow.**
A. ABI
B. API
C. HLL
D. ISA ✓
Answer: **D. ISA**

4 **The _____ defines the system call interface to the operating system and the hardware resources and services available in a system through the user instruction set architecture.**
A. HLL
B. API
C. ABI ✓
D. ISA
Answer: **C. ABI**

5 **The _____ gives a program access to the hardware resources and services available in a system through the user instruction set architecture supplemented with high-level language library calls.**

A. JCL
B. ISA
C. ABI
D. API ✓

Answer: **D. API**

6 **A _____ system works only one program at a time.**

A. batch
B. uniprogramming ✓
C. kernel
D. privileged instruction

Answer: **B. uniprogramming**

7 **A _____ is a special type of programming language used to provide instructions to the monitor.**

A. job control language ✓
B. multiprogram
C. kernel
D. utility

Answer: **A. job control language**

8 **The _____ scheduler determines which programs are admitted to the system for processing.**

A. long-term ✓
B. medium-term
C. short-term
D. I/O

Answer: **A. long-term**

9 **The _____ scheduler is also known as the dispatcher.**

A. long-term
B. medium-term
C. short-term ✓
D. I/O

Answer: **C. short-term**

10 **A _____ is an actual location in main memory.**

A. logical address
B. partition address
C. base address
D. physical address ✓

Answer: **D. physical address**

11 **_____ is when the processor spends most of its time swapping pages rather than executing instructions.**

A. Swapping
B. Thrashing ✓
C. Paging
D. Multitasking

Answer: **B. Thrashing**

12 **Virtual memory schemes make use of a special cache called a _____ for page table entries.**

A. TLB ✓
B. HLL
C. VMC
D. SPB

Answer: **A. TLB**

13 **With _____ the virtual address is the same as the physical address.**

A. unsegmented unpaged memory ✓
B. unsegmented paged memory
C. segmented unpaged memory
D. segmented paged memory

Answer: **A. unsegmented unpaged memory**

14 **A _____ is a collection of memory regions.**

A. APX
B. nucleus
C. domain ✓
D. page table

Answer: **C. domain**

15 **The OS maintains a _____ for each process that shows the frame location for each page of the process.**

A. kernel
B. page table ✓
C. TLB
D. logical address

Answer: **B. page table**

Number Systems

True / False — (15) صح / خطأ

1 **Our primary counting system is based on binary digits to represent numbers.**
Answer: **False**

2 **The decimal system has a radix of 100.**
Answer: **False**

3 **Negative powers of 10 are used to represent the positions of the numbers for decimal fractions.**
Answer: **True**

4 **A number with both an integer and fractional part has digits raised to both positive and negative powers of 10.**
Answer: **True**

5 **In any number, the rightmost digit is referred to as the most significant digit.**
Answer: **False**

6 **There are 50 tens in the number 509.**
Answer: **True**

7 **The decimal system is a special case of a positional number system with radix 10 and with digits in the range 0 through 9.**
Answer: **True**

8 **A number cannot be converted from binary notation to decimal notation.**
Answer: **False**

9 **Although convenient for computers, the binary system is exceedingly cumbersome for human beings.**
Answer: **True**

10 **A nibble is a grouping of four decimal digits.**
Answer: **False**

11 **Hexadecimal notation is only used for representing integers.**
Answer: **False**

12 **It is extremely easy to convert between binary and hexadecimal notation.**
Answer: **True**

13 **Hexadecimal notation is more compact than binary notation.**
Answer: **True**

14 **A sequence of hexadecimal digits can be thought of as representing an integer in base 10.**
Answer: **False**

15 **Because of the inherent binary nature of digital computer components, all forms of data within computers are represented by various binary codes.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **The decimal system has a base of _____.**
A. 0
B. 10 ✓
C. 100
D. 1000
Answer: **B. 10**

2 **Which digit represents "hundreds" in the number 8732?**
A. 8
B. 7 ✓
C. 3
D. 2
Answer: **B. 7**

3 **Which of the following is correct?**
A. 25 = (2 x 102) + (5 x 101)
B. 289 = (2 x 103) + (8 x 101) + (9 x 100)
C. 7523 = (7 x 103) + (5 x 102) + (2 x 101) + (3 x 100) ✓
D. 0.628 = (6 x 10-3) + (2 x 10-2) + (8 x 10-1)
Answer: **C. 7523 = (7 x 103) + (5 x 102) + (2 x 101) + (3 x 100)**

4 **In the number 3109, the 3 is referred to as the _____.**
A. most significant digit ✓
B. least significant digit
C. radix
D. base
Answer: **A. most significant digit**

5 **In the number 3109, the 9 is referred to as the _____.**

A. most significant digit
B. least significant digit ✓
C. radix
D. base

Answer: **B. least significant digit**

6 **Numbers in the binary system are represented to the _____.**

A. base 0
B. base 1
C. base 2 ✓
D. base 10

Answer: **C. base 2**

7 **Hexadecimal has a base of _____.**

A. 2
B. 8
C. 10
D. 16 ✓

Answer: **D. 16**

8 **The binary string 110111100001 is equivalent to _____.**

A. DE116 ✓
B. C7816
C. FF6416
D. B8F16

Answer: **A. DE116**

9 **The _____ system uses only the numbers 0 and 1.**

A. positional
B. binary ✓
C. hexadecimal
D. decimal

Answer: **B. binary**

10 **Decimal "10" is _____ in binary.**

A. 1000
B. 0010
C. 1010 ✓
D. 0001

Answer: **C. 1010**

11 **Decimal "10" is _____ in hexadecimal.**

A. 1
B. A ✓
C. 0
D. FF

Answer: **B. A**

12 **Four bits is called a _____.**

A. radix point
B. byte
C. nibble ✓
D. binary digit

Answer: **C. nibble**

13 **Another term for "base" is _____.**

A. radix ✓
B. integer
C. position
D. digit

Answer: **A. radix**

14 **In the number 472.156 the 2 is the _____.**

A. most significant digit
B. radix point
C. least significant digit
D. none of the above ✓

Answer: **D. none of the above**

15 **Binary 0101 is hexadecimal _____.**

A. 0
B. 5 ✓
C. A
D. 10

Answer: **B. 5**

Boolean Algebra & Digital Logic

True / False — (15) صح / خطأ

1 **The operation of the digital computer is based on the storage and processing of binary data.**
Answer: **True**

2 **Claude Shannon, a research assistant in the Electrical Engineering Department at M.I.T., proposed the basic principles of Boolean algebra.**
Answer: **False**

3 **In the absence of parentheses, the AND operation takes precedence over the OR operation.**
Answer: **True**

4 **Logical functions are implemented by the interconnection of decoders.**
Answer: **False**

5 **The delay by the propagation time of signals through the gate is known as the gate delay.**
Answer: **True**

6 **A combinational circuit consists of n binary inputs and m binary outputs.**
Answer: **True**

7 **Any Boolean function can be implemented in electronic form as a network of gates.**
Answer: **True**

8 **A Boolean function can be realized in the sum of products (SOP) form but not in the product of sums (POS) form.**
Answer: **False**

9 **"Don't care" conditions are when certain combinations of values of variables never occur, and therefore the corresponding output never occurs.**
Answer: **True**

10 **The value to be loaded into the program counter can come from a binary counter, the instruction register, or the output of the ALU.**
Answer: **True**

11 **In general, a decoder has n inputs and 2n outputs.**
Answer: **True**

12 **Combinational circuits are often referred to as "memoryless" circuits because their output depends only on their current input and no history of prior inputs is retained.**
Answer: **True**

13 **Binary addition is exactly the same as Boolean algebra.**
Answer: **False**

14 **Events in the digital computer are synchronized to a clock pulse so that changes occur only when a clock pulse occurs.**
Answer: **True**

15 **A register is a digital circuit used within the CPU to store one or more bits of data.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **The operand _____ yields true if and only if both of its operands are true.**
A. XOR
B. OR
C. AND ✓
D. NOT
Answer: **C. AND**

2 **The operation _____ yields true if either or both of its operands are true.**
A. NOT
B. AND
C. NAND
D. OR ✓
Answer: **D. OR**

3 **The unary operation _____ inverts the value of its operand.**
A. OR
B. NOT ✓
C. NAND
D. XOR
Answer: **B. NOT**

4 **A _____ is an electronic circuit that produces an output signal that is a simple Boolean operation on its input signals.**
A. gate ✓
B. decoder
C. counter
D. flip-flop
Answer: **A. gate**

5 **Which of the following is a functionally complete set?**

A. AND, NOT
B. NOR
C. AND, OR, NOT
D. all of the above ✓

Answer: **D. all of the above**

6 **For more than four variables an alternative approach is a tabular technique referred to as the _____ method.**

A. DeMorgan
B. Quine-McCluskey ✓
C. Karnaugh map
D. Boole-Shannon

Answer: **B. Quine-McCluskey**

7 **_____ are used in digital circuits to control signal and data routing.**

A. Multiplexers ✓
B. Program counters
C. Flip-flops
D. Gates

Answer: **A. Multiplexers**

8 **_____ is implemented with combinational circuits.**

A. Nano memory
B. Random access memory
C. Read only memory ✓
D. No memory

Answer: **C. Read only memory**

9 **The _____ exists in one of two states and, in the absence of input, remains in that state.**

A. assert
B. complex PLD
C. decoder
D. flip-flop ✓

Answer: **D. flip-flop**

10 **The _____ flip-flop has two inputs and all possible combinations of input values are valid.**

A. J-K ✓
B. D
C. S-R
D. clocked S-R

Answer: **A. J-K**

11 **A _____ accepts and/or transfers information serially.**

A. S-R latch
B. shift register ✓
C. FPGA
D. parallel register

Answer: **B. shift register**

12 **Counters can be designated as _____.**

A. asynchronous
B. synchronous
C. both asynchronous and synchronous ✓
D. neither asynchronous or synchronous

Answer: **C. both asynchronous and synchronous**

13 **CPUs make use of _____ counters, in which all of the flip-flops of the counter change at the same time.**

A. synchronous ✓
B. asynchronous
C. clocked S-R
D. timed ripple

Answer: **A. synchronous**

14 **The _____ table provides the value of the next output when the inputs and the present output are known, which is exactly the information needed to design the counter or any sequential circuit.**

A. excitation ✓
B. Kenough
C. J-K flip-flop
D. FPGA

Answer: **A. excitation**

15 **A _____ is a PLD featuring a general structure that allows very high logic capacity and offers more narrow logic resources and a higher ration of flip-flops to logic resources than do CPLDs.**

A. SPLD
B. FPGA ✓
C. PAL
D. PLA

Answer: **B. FPGA**

Instruction Sets: Characteristics & Functions

True / False — (15) صح / خطأ

1 **One boundary where the computer designer and the computer programmer can view the same machine is the machine instruction set.**
Answer: **True**

2 **The operation to be performed is specified by a binary code known as the operation code.**
Answer: **True**

3 **The address of the next instruction to be fetched must be a real address, not a virtual address.**
Answer: **False**

4 **It has become common practice to use a symbolic representation of machine instructions.**
Answer: **True**

5 **A high-level language expresses operations in a basic form involving the movement of data to or from registers.**
Answer: **False**

6 **One of the traditional ways of describing processor architecture is in terms of the number of addresses contained in each instruction.**
Answer: **True**

7 **Memory references are faster than register references.**
Answer: **False**

8 **The instruction set is the programmer's means of controlling the processor.**
Answer: **True**

9 **Addresses are a form of data.**
Answer: **True**

10 **Not all machine languages include numeric data types.**
Answer: **False**

11 **ARM processors support data types of 8 (byte), 16 (halfword), and 32 (word) bits in length.**
Answer: **True**

12 **Most machines provide the basic arithmetic operations of add, subtract, multiply, and divide.**
Answer: **True**

13 **A branch can be either forward or backward.**
Answer: **True**

14 **Procedures do not allow programming tasks to be subdivided into smaller units.**
Answer: **False**

15 **The focus of MMX technology is multimedia programming.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **The _____ specifies the operation to be performed.**
A. source operand reference
B. opcode ✓
C. next instruction reference
D. processor register
Answer: **B. opcode**

2 **A(n) _____ expresses operations in a concise algebraic form using variables.**
A. opcode
B. high-level language ✓
C. machine language
D. register
Answer: **B. high-level language**

3 **There must be _____ instructions for moving data between memory and the registers.**
A. branch
B. logic
C. memory ✓
D. I/O
Answer: **C. memory**

4 **_____ instructions operate on the bits of a word as bits rather than as numbers, providing capabilities for processing any other type of data the user may wish to employ.**
A. Logic ✓
B. Arithmetic
C. Memory
D. Test
Answer: **A. Logic**

5 **_____ instructions provide computational capabilities for processing number data.**

A. Boolean
B. Logic
C. Memory
D. Arithmetic ✓

Answer: **D. Arithmetic**

6 **_____ instructions are needed to transfer programs and data into memory and the results of computations back out to the user.**

A. I/O ✓
B. Transfer
C. Control
D. Branch

Answer: **A. I/O**

7 **The x86 data type that is a signed binary value contained in a byte, word, or doubleword, using twos complement representation is _____.**

A. general
B. ordinal
C. integer ✓
D. packed BCD

Answer: **C. integer**

8 **The most fundamental type of machine instruction is the _____ instruction.**

A. conversion
B. data transfer ✓
C. arithmetic
D. logical

Answer: **B. data transfer**

9 **The _____ instruction includes an implied address.**

A. skip ✓
B. rotate
C. stack
D. push

Answer: **A. skip**

10 **Which of the following is a true statement?**

A. a procedure can be called from more than one location
B. a procedure call can appear in a procedure
C. each procedure call is matched by a return in the called program
D. all of the above ✓

Answer: **D. all of the above**

11 **The entire set of parameters, including return address, which is stored for a procedure invocation is referred to as a _____.**

A. branch
B. stack frame ✓
C. pop
D. push

Answer: **B. stack frame**

12 **Which ARM operation category includes logical instructions (AND, OR, XOR), add and subtract instructions, and test and compare instructions?**

A. data-processing instructions ✓
B. branch instructions
C. load and store instructions
D. extend instructions

Answer: **A. data-processing instructions**

13 **In the ARM architecture only _____ instructions access memory locations.**

A. data processing
B. status register access
C. load and store ✓
D. branch

Answer: **C. load and store**

14 **Which data type is defined in MMX?**

A. packed byte
B. packed word
C. packed doubleword
D. all of the above ✓

Answer: **D. all of the above**

15 **A branch instruction in which the branch is always taken is _____.**

A. conditional branch
B. unconditional branch ✓
C. jump
D. bi-endian

Answer: **B. unconditional branch**

Instruction Sets: Addressing Modes & Formats

True / False — (15) صح / خطأ

1 **The value of the mode field determines which addressing mode is to be used.**
Answer: **True**

2 **In a system without virtual memory, the effective address is a virtual address or a register.**
Answer: **False**

3 **The disadvantage of immediate addressing is that the size of the number is restricted to the size of the address field.**
Answer: **True**

4 **With direct addressing, the length of the address field is usually less than the word length, thus limiting the address range.**
Answer: **True**

5 **Register addressing is similar to direct addressing with the only difference being that the address field refers to a register rather than a main memory address.**
Answer: **True**

6 **Register indirect addressing uses the same number of memory references as indirect addressing.**
Answer: **False**

7 **Three of the most common uses of stack addressing are relative addressing, base-register addressing, and indexing.**
Answer: **False**

8 **The method of calculating the EA is the same for both base-register addressing and indexing.**
Answer: **True**

9 **Typically an instruction set will include both preindexing and postindexing.**
Answer: **False**

10 **The x86 is equipped with a variety of addressing modes intended to allow the efficient execution of high-level languages.**
Answer: **True**

11 **The base with index and displacement mode sums the contents of the base register, the index register, and a displacement to form the effective address.**
Answer: **True**

12 **The memory transfer rate has not kept up with increases in processor speed.**
Answer: **True**

13 **For addresses that reference memory the range of addresses that can be referenced is not related to the number of address bits.**
Answer: **False**

14 **The principal price to pay for variable-length instructions is an increase in the complexity of the processor.**
Answer: **True**

15 **One advantage of linking the addressing mode to the operand rather than the opcode is that any addressing mode can be used with any opcode.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **The advantage of _____ is that no memory reference other than the instruction fetch is required to obtain the operand.**

A. direct addressing
B. immediate addressing ✓
C. register addressing
D. stack addressing

Answer: **B. immediate addressing**

2 **The principal advantage of _____ addressing is that it is a very simple form of addressing.**

A. displacement
B. register
C. stack
D. direct ✓

Answer: **D. direct**

3 **_____ has the advantage of large address space, however it has the disadvantage of multiple memory references.**

A. Indirect addressing ✓
B. Direct addressing
C. Immediate addressing
D. Stack addressing

Answer: **A. Indirect addressing**

4 **The advantages of _____ addressing are that only a small address field is needed in the instruction and no time-consuming memory references are required.**

A. direct
B. indirect
C. register ✓
D. displacement

Answer: **C. register**

5 **_____ has the advantage of flexibility, but the disadvantage of complexity.**

A. Stack addressing
B. Displacement addressing ✓
C. Direct addressing
D. Register addressing

Answer: **B. Displacement addressing**

6 **For _____, the address field references a main memory address and the referenced register contains a positive displacement from that address.**

A. indexing ✓
B. base-register addressing
C. relative addressing
D. all of the above

Answer: **A. indexing**

7 **Indexing performed after the indirection is _____.**

A. relative addressing
B. autoindexing
C. postindexing ✓
D. preindexing

Answer: **C. postindexing**

8 **For the _____ mode, the operand is included in the instruction.**

A. immediate ✓
B. base
C. register
D. displacement

Answer: **A. immediate**

9 **The only form of addressing for branch instructions is _____ addressing.**

A. register
B. relative
C. base
D. immediate ✓

Answer: **D. immediate**

10 **Which of the following interrelated factors go into determining the use of the addressing bits?**

A. number of operands
B. number of register sets
C. address range
D. all of the above ✓

Answer: **D. all of the above**

11 **_____ is a principle by which two variables are independent of each other.**

A. Opcode
B. Orthogonality ✓
C. Completeness
D. Autoindexing

Answer: **B. Orthogonality**

12 **The _____ was designed to provide a powerful and flexible instruction set within the constraints of a 16-bit minicomputer.**

A. PDP-1
B. PDP-8
C. PDP-11 ✓
D. PDP-10

Answer: **C. PDP-11**

13 **The _____ byte consists of three fields: the Scale field, the Index field and the Base field.**

A. SIB ✓
B. VAX
C. PDP-11
D. ModR/M

Answer: **A. SIB**

14 **All instructions in the ARM architecture are _____ bits long and follow a regular format.**

A. 8
B. 16
C. 32 ✓
D. 64

Answer: **C. 32**

15 **_____ is a design principle employed in designing the PDP-10 instruction set.**

A. Orthogonality
B. Completeness
C. Direct addressing
D. All of the above ✓

Answer: **D. All of the above**

Processor Structure & Function, Pipelining

True / False — (15) صح / خطأ

1 **The processor needs to store instructions and data temporarily while an instruction is being executed.**
Answer: **True**

2 **The control unit (CU) does the actual computation or processing of data.**
Answer: **False**

3 **Within the processor there is a set of registers that function as a level of memory above main memory and cache in the hierarchy.**
Answer: **True**

4 **Condition codes facilitate multiway branches.**
Answer: **True**

5 **The allocation of control information between registers and memory are not considered to be a key design issue.**
Answer: **False**

6 **Instruction pipelining is a powerful technique for enhancing performance but requires careful design to achieve optimum results with reasonable complexity.**
Answer: **True**

7 **The cycle time of an instruction pipeline is the time needed to advance a set of instructions one stage through the pipeline.**
Answer: **True**

8 **A control hazard occurs when two or more instructions that are already in the pipeline need the same resource.**
Answer: **False**

9 **One of the major problems in designing an instruction pipeline is assuring a steady flow of instructions to the initial stages of the pipeline.**
Answer: **True**

10 **The predict-never-taken approach is the most popular of all the branch prediction methods.**
Answer: **True**

11 **It is possible to improve pipeline performance by automatically rearranging instructions within a program so that branch instructions occur later than actually desired.**
Answer: **True**

12 **Interrupt processing allows an application program to be suspended in order that a variety of interrupt conditions can be serviced and later resumed.**
Answer: **True**

13 **An interrupt is generated from software and it is provoked by the execution of an instruction.**
Answer: **False**

14 **While the processor is in user mode the program being executed is unable to access protected system resources or to change mode, other than by causing an exception to occur.**
Answer: **True**

15 **The exception modes have full access to system resources and can change modes freely.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **_____ are a set of storage locations.**
A. Processors
B. PSWs
C. Registers ✓
D. Control units
Answer: **C. Registers**

2 **The _____ controls the movement of data and instructions into and out of the processor.**
A. control unit ✓
B. ALU
C. shifter
D. branch
Answer: **A. control unit**

3 **_____ registers may be used only to hold data and cannot be employed in the calculation of an operand address.**
A. General purpose
B. Data ✓
C. Address
D. Condition code
Answer: **B. Data**

4 **_____ are bits set by the processor hardware as the result of operations.**
A. MIPS
B. Condition codes ✓
C. Stacks
D. PSWs
Answer: **B. Condition codes**

5 **The _____ contains the address of an instruction to be fetched.**

A. instruction register
B. memory address register
C. memory buffer register
D. program counter ✓

Answer: **D. program counter**

6 **The _____ contains a word of data to be written to memory or the word most recently read.**

A. MAR
B. PC
C. MBR ✓
D. IR

Answer: **C. MBR**

7 **The _____ determines the opcode and the operand specifiers.**

A. decode instruction ✓
B. fetch operands
C. calculate operands
D. execute instruction

Answer: **A. decode instruction**

8 **_____ is a pipeline hazard.**

A. Control
B. Resource
C. Data
D. All of the above ✓

Answer: **D. All of the above**

9 **A _____ hazard occurs when there is a conflict in the access of an operand location.**

A. resource
B. data ✓
C. structural
D. control

Answer: **B. data**

10 **A _____ is a small, very-high-speed memory maintained by the instruction fetch stage of the pipeline and containing the n most recently fetched instructions in sequence.**

A. loop buffer ✓
B. delayed branch
C. multiple stream
D. branch prediction

Answer: **A. loop buffer**

11 **The _____ is a small cache memory associated with the instruction fetch stage of the pipeline.**

A. dynamic branch
B. loop table
C. branch history table ✓
D. flag

Answer: **C. branch history table**

12 **The _____ stage includes ALU operations, cache access, and register update.**

A. decode
B. execute ✓
C. fetch
D. write back

Answer: **B. execute**

13 **_____ is used for debugging.**

A. Direction flag
B. Alignment check
C. Trap flag ✓
D. Identification flag

Answer: **C. Trap flag**

14 **The ARM architecture supports _____ execution modes.**

A. 2
B. 8
C. 11
D. 7 ✓

Answer: **D. 7**

15 **The OS usually runs in _____.**

A. supervisor mode ✓
B. abort mode
C. undefined mode
D. fast interrupt mode

Answer: **A. supervisor mode**

RISC (Reduced Instruction Set Computers)

True / False — (15) صح / خطأ

1 **Microprogramming eases the task of designing and implementing the control unit and provides support for the family concept.**
Answer: **True**

2 **Pipelining is a means of introducing parallelism into the essentially sequential nature of a machine-instruction program.**
Answer: **True**

3 **The major cost in the life cycle of a system is hardware.**
Answer: **False**

4 **It is common for programs, both system and application, to continue to exhibit new bugs after years of operation.**
Answer: **True**

5 **Procedure calls and returns are not important aspects of HLL programs.**
Answer: **False**

6 **The register file is on the same chip as the ALU and control unit.**
Answer: **True**

7 **The register file employs much shorter addresses than addresses for cache and memory.**
Answer: **True**

8 **To handle any possible pattern of calls and returns the number of register windows would have to be unbounded.**
Answer: **True**

9 **Cache memory is a much faster memory than the register file.**
Answer: **False**

10 **The cache is capable of handling global as well as local variables.**
Answer: **True**

11 **When using graph coloring, nodes that share the same color cannot be assigned to the same register.**
Answer: **False**

12 **With simple, one cycle instructions, there is little or no need for microcode.**
Answer: **True**

13 **Almost all RISC instructions use simple register addressing.**
Answer: **True**

14 **RISC processors are more responsive to interrupts because interrupts are checked between rather elementary operations.**
Answer: **True**

15 **Unrolling can improve performance by increasing instruction parallelism by improving pipeline performance.**
Answer: **True**

Multiple Choice — (15) اختيار من متعدد

1 **_____ determines the control and pipeline organization.**
A. Calculation
B. Execution sequencing ✓
C. Operations performed
D. Operands used
Answer: **B. Execution sequencing**

2 **The Patterson study examined the dynamic behavior of _____ programs, independent of the underlying architecture.**
A. HLL ✓
B. RISC
C. CISC
D. all of the above
Answer: **A. HLL**

3 **_____ is the fastest available storage device.**
A. Main memory
B. Cache
C. Register storage ✓
D. HLL
Answer: **C. Register storage**

4 **The first commercial RISC product was _____.**
A. SPARC
B. CISC
C. VAX
D. the Pyramid ✓
Answer: **D. the Pyramid**

5 **_____ instructions are used to position quantities in registers temporarily for computational operations.**

A. Load-and-store ✓
B. Window
C. Complex
D. Branch

Answer: **A. Load-and-store**

6 **Which stage is required for load and store operations?**

A. I
B. E
C. D
D. all of the above ✓

Answer: **D. all of the above**

7 **A _____ instruction can be used to account for data and branch delays.**

A. SUB
B. NOOP ✓
C. JUMP
D. all of the above

Answer: **B. NOOP**

8 **The instruction location immediately following the delayed branch is referred to as the _____.**

A. delay load
B. delay file
C. delay slot ✓
D. delay register

Answer: **C. delay slot**

9 **A tactic similar to the delayed branch is the _____, which can be used on LOAD instructions.**

A. delayed load ✓
B. delayed program
C. delayed slot
D. delayed register

Answer: **A. delayed load**

10 **The MIPS R4000 uses _____ bits for all internal and external data paths and for addresses, registers, and the ALU.**

A. 16
B. 32
C. 64 ✓
D. 128

Answer: **C. 64**

11 **All MIPS R series processor instructions are encoded in a single _____ word format.**

A. 4-bit
B. 8-bit
C. 16-bit
D. 32-bit ✓

Answer: **D. 32-bit**

12 **A _____ architecture is one that makes use of more, and more fine-grained pipeline stages.**

A. parallel
B. superpipelined ✓
C. superscalar
D. hybrid

Answer: **B. superpipelined**

13 **The R4000 can have as many as _____ instructions in the pipeline at the same time.**

A. 8 ✓
B. 10
C. 5
D. 3

Answer: **A. 8**

14 **SPARC refers to an architecture defined by _____.**

A. Microsoft
B. Apple
C. Sun Microsystems ✓
D. IBM

Answer: **C. Sun Microsystems**

15 **The R4000 pipeline stage where the instruction result is written back to the register file is the _____ stage.**

A. write back ✓
B. tag check
C. data cache
D. instruction execute

Answer: **A. write back**